INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II , Third Semester Examination, October-November 2026

Subject Code: BCACC303

Subject Name: Artificial Intelligence using Python

Full Marks: 30 Set No:2 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. Mention any two advantages of Python for data analytics. **2**

2. What is an exception in Python? Give one example of a standard exception. **2**

3. Differentiate between a text file and a binary file in Python. **2**

4. What is learning rate (α)? **2**

5. State from where Rosenblatt's perceptron model is adapted. **2**

6. Give two examples of informed search techniques. **2**

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Describe the main file modes used to read, write, and update files in Python. Explain why files should be closed after use. **5**

8. Explain the purpose of try, except, and finally blocks in Python exception handling. **5**

9. Briefly explain DFS algorithm with proper example. **5**

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Explain the data analytics workflow and the roles of Pandas, NumPy, Matplotlib, and Seaborn at different stages. Illustrate how these libraries work together to turn raw data into useful insights. **10**

11. Explain back propagation networks. **10**

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Full Marks: 30 Set No:3 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. Differentiate between a text file and a binary file in Python. **2**

2. What is an exception in Python? Give one example of a standard exception. **2**

3. What is the difference between crisp logic and fuzzy logic? **2**

4. What is search in AI? **2**

5. Give two examples of uninformed search techniques. **2**

6. Who invented ADALINE? **2**

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Explain how Python can handle multiple exceptions using except clauses. Include a suitable example or pseudocode. **5**

8. Explain the purpose of try, except, and finally blocks in Python exception handling. **5**

9. Explain different neural network architectures using proper diagrams. **5**

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Explain the steps involved in data analytics using Python. Describe the uses of NumPy, Pandas, Matplotlib, and Seaborn in data analysis and visualization. Give a suitable example to show how these libraries help analyze data and present the results. **10**

11. What is heuristic search? State advantages and limitations of heuristic search. **10**

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Subject Name: Artificial Intelligence using Python

Full Marks: 30 Set No:1 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Define a fuzzy set and state the range of a membership function. **2**

2. What is the purpose of the **with** statement (context manager) when handling files in Python? **2**

3. Differentiate between a text file and a binary file in Python. **2**

4. What is a perceptron? **2**

5. What is an activation function? **2**

6. Give two examples of blind search. **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Explain the five steps in a data analytics pipeline: collect, clean, analyze, visualize, and decide **5**

8. Explain the purpose of try, except, and finally blocks in Python exception handling. **5**

9. Briefly explain DFS algorithm with proper example. **5**

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Explain the data analytics workflow using Python. Discuss the roles of NumPy, Pandas, Matplotlib, and Seaborn in data collection, cleaning, analysis, and visualization. Illustrate how these libraries work together to extract meaningful insights from a dataset with a suitable example. **10**

11. Describe Rosenblatt's perceptron model. **10**